EKAMVEER **SINGH**

Chandigarh 160101 | +91 9530676868 | ekamveer.singh06@gmail.com | github.com/Cheese-Singh

SUMMARY

Third-year B.Tech student in Computer Science and Artificial Intelligence with a focus on efficient on-device ML inference and privacy-preserving AI systems. Building production-grade local AI pipelines from scratch on Apple Silicon, progressing from low-level tensor operations to a working voice-to-LLM assistant pipeline. Actively contributing to interdisciplinary AI research alongside PhD scholars and faculty.

EDUCATION

B.Tech — Computer Science and Artificial Intelligence (CGPA: 9.01) Expected May 2028
Punjab Engineering College – Chandigarh, India

Minor — Foundations of Quantum Technology (CGPA: 10.00) Expected May 2028
Punjab Engineering College – Chandigarh, India

SKILLS

Languages: Python, C++, Swift, SQL
ML / AI: MLX, MLX-VLM, PyTorch, TensorFlow, Scikit-learn, CatBoost, MLX-Whisper, OpenCV
On-Device & Systems: MLX Swift, SwiftUI, Apple Silicon inference, MLX-audio, sounddevice
Data & Visualisation: NumPy, Pandas, Matplotlib, Seaborn, Plotly, MATLAB
Tools & Infra: Git, GitHub, Docker, VSCode, AWS, Ollama, FastAPI, Kaggle API, OpenAlex API
Databases: MySQL, PostgreSQL, MongoDB, SQLite

PROJECTS

Gwen — Local Personal AI Assistant April 2026-Present
github.com/Cheese-Singh - [MLX Series — Side Projects]()

- Building a fully local, privacy-first voice AI assistant for Apple Silicon (MacBook M3, iPhone, Apple Watch), combining on-device inference with optional cloud fallback for complex queries.
- Architected a 4-tier model routing system (Qwen2.5-1.5B → Qwen3.5-9B → Gemma4b:cloud → Kimi-K2.7-Code:cloud via Ollama Cloud) with automatic cloud-to-local fallback, balancing latency, cost, and response quality.
- Implemented speaker verification and voice authentication using an ECAPA-TDNN model (SpeechBrain), and acoustic echo cancellation (WebRTC AEC3) for reliable full-duplex, barge-in-capable conversation.
- Built multimodal I/O: local image generation (ZImageTurbo, 4-bit), image/PDF/DOCX understanding and generation, and a novel query-routing system for grounding responses and reducing hallucination.
- Progressive 15-stage pipeline from tensor fundamentals and manual embeddings (A1) through full voice-driven system agent (A10) to speaker-verified, multi-tier, multimodal assistant (A15).
- Stack: mlx, mlx-lm, mlx-vlm, mlx-whisper, mlx-audio (Kokoro-82M), speechbrain, ollama, sounddevice, mflux. All code written independently; every stage committed separately.

Automated Fungal Keratitis Diagnosis using Machine Learning July 2026
github.com/MatishaKansal/Automated-Fungal-Keratitis-Classifier - [PGI-MER Project]()

- Collaborated in a 3-person team to build an end-to-end computer vision system for screening fungal keratitis, a severe corneal infection, using In Vivo Confocal Microscopy images and patient risk-factor history, in partnership with PGI-MER
- Led system architecture and planning, and built the LLM-based summary generation layer that translates model outputs and patient risk factors into clinician-readable diagnostic summaries
- Contributed to a stacked ensemble classifier (Mixture of Experts plus Snapshot Ensemble of Random Forests) benchmarked across seven image-processing pipelines, reaching approximately 99 to 100 percent accuracy on held-out test data
- Stack: Python, FastAPI, scikit-learn, OpenCV, React, LLM integration (Ollama)

Local LLM Inference & Agentic Tool-Use Series March 2026
github.com/Cheese-Singh — [Ollama Series — Side Projects]()

- Built a 5-script progressive series covering single-turn generation, multimodal vision-language input, stateful multi-turn chat with think-block stripping and repetition prevention, and an agentic function-calling loop.
- Agentic script (B4) implements a full tool-use loop: LLM reasons over a CSV-backed inventory, calls structured functions (check, add, update, save), handles errors, and executes multi-step plans autonomously.
- Final script (B5) builds a CLI for Modelfile lifecycle management, model validation, and automated prompt benchmarking with structured report generation.

- Stack: ollama, pandas, Python.

Research ML — Materials Science Prediction & Trend Analysis 2025

github.com/Cheese-Singh — Research ML — Side Projects

- Built a CatBoost regression pipeline to predict supercapacitor capacitance (F/g) from material, synthesis, and electrolyte features; evaluated with Leave-One-Out Cross-Validation and generated diagnostic plots (actual-vs-predicted, residual distribution, feature importance).
- Built a research trend analyser querying the OpenAlex API to track yearly publication volume for any topic and output bar chart visualisations.
- Stack: CatBoost, scikit-learn, pandas, NumPy, Matplotlib, OpenAlex API.

ML Foundations — Classical ML from Scratch Nov 2025-2026

github.com/Cheese-Singh — ML Foundations Repository

- A structured series of standalone scripts each isolating one classical ML concept: multi-classifier comparison (KNN, SVM, Decision Tree), SVM decision boundary visualisation across kernels and C values, automated EDA pipeline generating correlation heatmaps and histograms for any CSV.
- Includes a full sklearn preprocessing pipeline (Pipeline + ColumnTransformer) with imputation, scaling, and encoding — zero manual cleaning — demonstrated on the Titanic dataset.
- PCA vs. t-SNE interactive comparison on Fashion-MNIST using Plotly, highlighting the difference between linear and nonlinear dimensionality reduction.

RESEARCH & PUBLICATIONS

Co-author — "Interface-Driven Electrochemical Performance of Hybrid MXenes for High-Performance Supercapacitors: Insights and Challenges"

Submitted to Elsevier (2026). Interdisciplinary review written with a PhD scholar under faculty supervision. Independently authored the AI methodology section, building ML regression models to analyse electrochemical performance data.

EXPERIENCE

Research Intern Mar 2025 – Jul 2025

Punjab Engineering College – Chandigarh, India

- Conducted literature reviews for an interdisciplinary AI + materials science project spanning supercapacitor electrochemistry and ML-based performance prediction.
- Independently authored the AI methodology section of a co-authored review article submitted to Elsevier; implemented CatBoost regression pipeline with LOOCV evaluation.
- Collaborated with a PhD scholar and faculty from Computer Science and Physics departments.

ACCOMPLISHMENTS

- Finalist, Amazon ML Summer School 2026
- JEE Mains 2024: 98.8 percentile
- Published: "Solar-Powered Mobile Phones: The Future of Sustainable Energy", The Tribune (2023)
- Third Place, Intra-college Robosoccer Competition
- Participant, Smart India Hackathon (SIH) 2024

CLUBS & ACTIVITIES

- Technical Society Member — ACM & Robotics Club (Aug 2024 – Present)
- Counselling Cell Member (Aug 2024 – Present)
- NGO Volunteer — Aashray Foundation (Mar – Dec 2025)
- Football Team Player

LANGUAGES

Punjabi (Native) **English** (C2 — Proficient) **Hindi** (C2 — Proficient) **Norwegian** (A1 — Beginner)